

were Stephen Wiesner, Charles Bennett and Gilles Brassard. They showed how a provable secure mechanism is provided by the non-classical properties of quantum measurement for critical cryptographic establishment. This was also recognised by Richard Feynman and Yuri Manin, along with other scientists — that the Turing machine could not simulate a few quantum phenomena related to entangled particles. This reflection led to the hypothesis that possibly, these quantum processes could be used to accelerate general computation. This hypothesis would require rethinking the model of information theory underlying computation. It meant taking the model out of the pure classical realm and placing it in the quantum realm.

The early 1980s saw the beginning of quantum computing with physicist Paul Benioff's proposal of a quantum-mechanical model of the Turing machine in his paper published in the Journal of Statistical Physics. Feynman and Manin later shared the opinion that quantum computing has the potency of simulating classically impossible things. David Deutsch, an Oxford University theoretical physicist who is considered the father of quantum computing, suggested the idea of the quantum mechanical Turing machine. Later, Ethan Bernstein and Umesh Vazirani studied his model intensely and demonstrated that a quantum Turing machine is capable of simulating a classical Turing machine by improving on it. So, it was evident that any classical computation could be simulated quantumly with, at most, a polynomial-time slowdown.

Quantum computing explores the implications of replacing classical mechanics with quantum mechanics for information modelling and processing. It is not about replacing the physical substrate of computation but rather changing the notion of computation itself from classical to quantum. The conventional computers we work with on a daily basis use quantum mechanics but computation is done using

Table 1: The differences between classical and quantum computing

Classical computing	Quantum computing
Its computations and circuit behaviour are based on classical physics.	Its computations are based on quantum physics.
Information storage is in bits based on voltage/charge, etc.	Information storage is in qubits based on electron-spin direction.
Bits can be either 0 or 1 at a time.	Qubits, at any time, are a combination of 0 and 1. They can hold many complex values and even negative values.
Deterministic approach: Repeated computations on the same input will lead to the same output.	Probabilistic approach: Measurements on super-positioned states give probabilistic answers and are then reduced to 0 or 1.
Logic gates are used for information processing - AND, OR, NOT, etc, in a sequential manner.	Quantum logic gates are used for information processing in a parallel manner.
A discrete number of possible states is possible: 0 or 1.	An infinite number of possible states and qubits can have continuous values.
Operations are defined as per Boolean algebra.	Operations are defined based on linear algebra over Hilbert spaces.

classical bits. In this quantum notion of computation, these bits are replaced by qubits (i.e., quantum bits).

The 'quantum' in quantum computing

The fundamental unit of information in the world of quantum computing is known as a qubit. We can imagine this qubit as an electron located in

a magnetic field, where its state can be either lined-up with the field (i.e., spin-up or 0) or not lined-up with the field (i.e., spin-down or 1) (Figure 1). The electron's state is changed using an energy pulse. Each qubit is capable of being in more than one state at any given point in time and hence can take superposition of both 0 and 1. A quantum computer can take 2^n

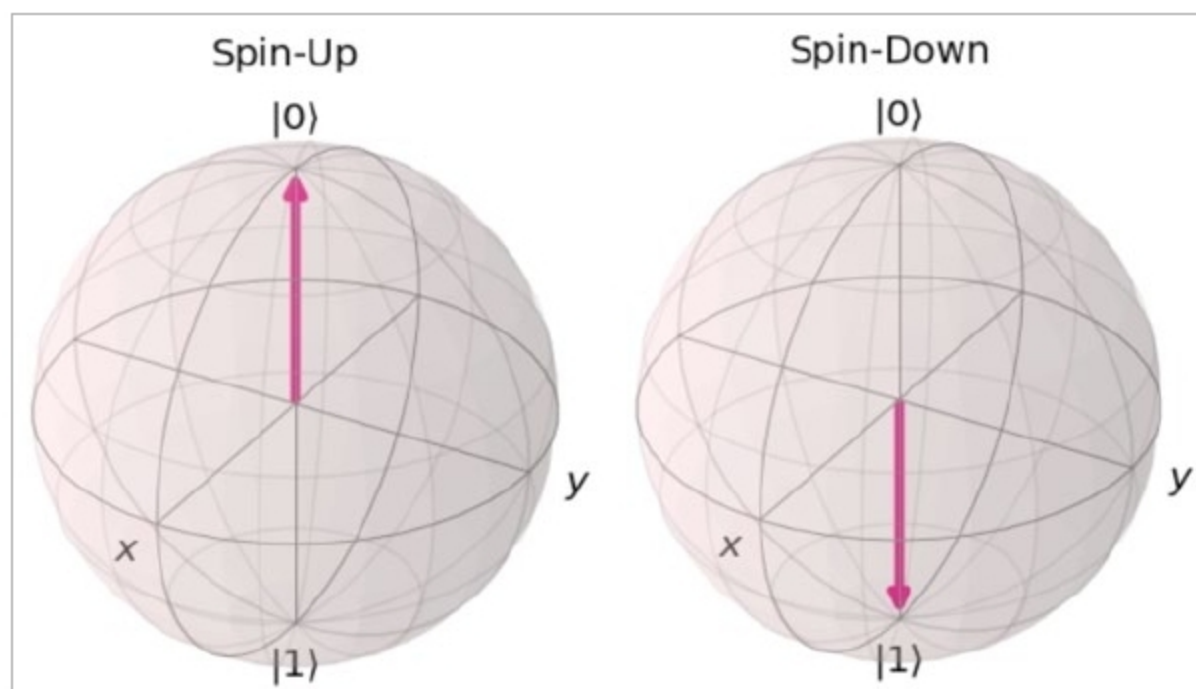


Figure 1: Spin-up and spin-down states of a qubit visualised programmatically using the Qiskit-Python library